

Dynamic Macroeconomics II

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Optimal Fiscal Policy with Commitment

Second Part

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Before we continue...

- Recall that to make Ramsey Problem interesting we rule out lump-sum taxes
- Why? Those taxes do not affect marginal decisions, i.e. they are not distortionary
- We can think of Ramsey Problem as minimizing distortions caused by taxes that finance a given sequence of government spending.
- If lump-sum taxes were available as a policy tool, it would be Ramsey-optimal to use them and set all distortionary taxes equal to zero.
- That is not an interesting solution.
- In real world governments use distortionary taxes.

Why is it Ramsey-optimal to have $\tau_k = 0$?

... at least in the neoclassical growth model?

- Capital is the reproducible factor of production.
- More investment today yields higher output and consumption in the future, for both (of Judd's) capitalists and workers
- Marginal product of labor increases with capital/labor, i.e., $w(K/L)$
- Therefore accumulation of capital increases real wages, benefitting workers.

Primal approach to Ramsey Problem

- Chamley eliminated taxes from Problem.
- Primal approach: now, we eliminate all prices and taxes.
- Government chooses a feasible allocation subject to constraints that ensure the existence of prices and taxes such that the chosen allocation is consistent with a competitive equilibrium.
particular

- We use same economy as with Chamley; government can issue debt.
- We will be using the intertemporal budget constraint.
- As you know, to construct it you use sequential constraints, eliminating debt:

$$c_t + \frac{c_{t+1}}{R_t} + \frac{k_{t+2}}{R_t} + \frac{b_{t+2}}{R_t R_{t+1}} = b_t + \dots$$

- Iterate.
- All terms related to the choice of capital tomorrow disappear as we impose the No-arbitrage Condition, which you can find.
- You can find the Transversality Conditions:

$$\lim_{T \rightarrow \infty} \left(\prod_{i=0}^{T-1} R_i^{-1} \right) k_{T+1} = 0$$

$$\lim_{T \rightarrow \infty} \left(\prod_{i=0}^{T-1} R_i^{-1} \right) \frac{b_{T+1}}{R_T} = 0$$

- The household's present-value budget constraint is:

$$\sum_{t=0}^{\infty} q_t c_t = \sum_{t=0}^{\infty} q_t (1 - \tau_{nt}) w_t n_t + [(1 - \tau_{k0}) r_0 + 1 - \delta] k_0 + b_0$$

Notice I have imposed No Arbitrage Condition, eliminating k_{t+1} terms

- q_t is the Arrow-Debreu price:

$$q_t = \prod_{i=0}^{t-1} R_i^{-1}, t \geq 1$$

with $q_0 = 1$.

Steps of Primal approach

Step 0: what we just did, construct intertemporal budget constraint, imposing NAC. Then:

- ① **A.** Keep track of Euler equation/No-arbitrage condition from household, either from sequential problem or from intertemporal constraint, respectively. $\rightarrow Z \cdot U_1(\cdot)$
B. Calculate first-order conditions for household's problem and firm's problem. Use FOCs to solve for prices and taxes as functions of allocations.
- ② Substitute these expressions into the household's present-value budget constraint. This is how we get the Implementability Condition (IC).
- ③ Solve Ramsey Problem by maximizing household's utility subject to feasibility and IC.
- ④ Use formulas from Step 1 to compute prices and taxes.

Step 1

- Let λ be Lagrange multiplier for intertemporal budget constraint. Household's problem first-order conditions:

$$c_t : \beta^t u_{ct} = \lambda q_t$$

$$n_t : \beta^t u_{lt} = \lambda q_t (1 - \tau_{nt}) w_t$$

which imply

$$q_t = \beta^t \frac{u_{ct}}{u_{c0}}$$

$$(1 - \tau_{nt}) w_t = \frac{u_{lt}}{u_{ct}}$$

- First equation comes from

$$q_t = \beta^t \frac{u_{ct}}{\lambda}$$

evaluated at $t = 0$ which implies, with $q_0 = 1$, that $\lambda = u_{c0}$.

- Take Euler equation/No-arbitrage Condition. Originally

$$R_t = (1 - \tau_{kt+1})r_{t+1} + 1 - \delta$$

- Notice that we can write it using intertemporal prices as

$$\frac{q_t}{q_{t+1}} = (1 - \tau_{kt+1})r_{t+1} + 1 - \delta$$

- From firm's problem:

$$F_{kt} = r_t$$

$$F_{nt} = w_t$$

Step 2

- Substitute FOCs and firm's input prices into intertemporal budget constraint (**check**):

$$\sum_{t=0}^{\infty} \beta^t \frac{u_{ct}}{u_{c0}} c_t = \sum_{t=0}^{\infty} \beta^t \frac{u_{ct}}{u_{c0}} \frac{u_{lt}}{u_{ct}} n_t + \{[(1 - \tau_{k0})F_{k0} + 1 - \delta]k_0 + b_0\} = 0$$

- Define

$$A \equiv u_{c0} \{[(1 - \tau_{k0})F_{k0} + 1 - \delta]k_0 + b_0\}$$

- Multiply constraint by u_{c0} to get **Implementability Condition**:

$$\sum_{t=0}^{\infty} \beta^t (u_{ct} c_t - u_{lt} n_t) - A = 0$$

Step 3

- Maximize utility subject to IC and feasibility.
- Let Φ be a Lagrange multiplier and define:

$$V(c_t, n_t, \Phi) \equiv u(c_t, 1 - n_t) + \Phi(u_c(t)c_t - u_l(t)n_t)$$

- Note that the discounted sum over t of this expression is equivalent to maximizing utility subject to IC, except for A .
- Write the Lagrangian:

$$\begin{aligned} \mathcal{J} = & \sum_{t=0}^{\infty} \beta^t \{ V(c_t, n_t, \Phi) + \\ & \theta_t [F(k_t, n_t) - c_t - k_{t+1} + (1 - \delta)k_t - g_t] \} \\ & - \Phi A \end{aligned}$$

where θ_t is the multiplier for feasibility.

- Given k_0 and b_0 , fix τ_{k_0} and maximize $\mathcal{J}$ with respect to $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$.

- First-order conditions:

$$c_t : V_{ct} = \theta_t$$

$$n_t : V_{nt} = -\theta_t F_{nt}$$

$$k_{t+1} : \theta_t = \beta \theta_{t+1} [F_{kt+1} + 1 - \delta]$$

$$c_0 : V_{c0} = \theta_0 + \Phi A_c$$

$$n_0 : V_{n0} = -\theta_0 F_{n0} + \Phi A_n.$$

- The first-order condition for capital becomes:

$$V_{ct} = \beta V_{c,t+1} [F_{kt+1} + (1 - \delta)].$$

- We also have the FOC with respect to multipliers:

$$\theta_t : F(k_t, n_t) - c_t - k_{t+1} + (1 - \delta)k_t - g_t = 0$$

$$\Phi : \sum_{t=0}^{\infty} \beta^t (u_{ct} c_t - u_{lt} n_t) - A = 0$$

- We seek an allocation $c_0, n_0, \{c_t, n_t, k_t\}_{t=1}^{\infty}$ and a multiplier Φ ; we have 6 equations and 6 unknowns.

Step 4

- Recall that the main question here is to find the Ramsey-optimal τ_k ; let us focus on that.
- Assume the solution converges to a steady-state.
- FOC for capital becomes:

$$1 = \beta[F_k + (1 - \delta)]$$

- We kept track of the NAC:

$$\frac{q_t}{q_{t+1}} = (1 - \tau_{kt+1})F_{kt+1} + 1 - \delta$$

- In a steady state (**check**)

$$1/\beta = (1 - \tau_k)F_k + 1 - \delta$$

- Comparing the two, the implication is that $\tau_k = 0$.

- The nonnegative Lagrange multiplier Φ measures the utility cost of raising government revenue through distorting taxes.
- If $\Phi > 0$, it reflects the welfare cost of the distorted margins.
- Distorting taxes are putting an extra constraint on the economy.
- To put it differently, notice that $\Phi = 0$ makes the Ramsey Problem equivalent to maximizing utility subject to feasibility, which is the solution to the Planner's Problem, and to an undistorted Competitive Equilibrium.

$$RPI, CNA + CPO = C1$$

An example where $\tau_k = 0$ breaks down: Incomplete taxation

- Correia (1996) introduces an additional production factor z_t , which is inelastically supplied ($z_t = Z$).
- Production function $F(k_t, n_t, z_t)$: constant returns to scale, strictly positive decreasing marginal returns
- Profit maximization: Price of factor z_t is $p_{z_t} = F_{z_t}$.
- Add a stream of revenues to household's intertemporal budget constraint:

$$\sum_{t=0}^{\infty} q_t p_{z_t} Z$$

- Implementability condition (**check**):

$$\sum_{t=0}^{\infty} \beta^t \frac{u_{c_t}}{u_{c_0}} c_t = \sum_{t=0}^{\infty} \beta^t \frac{u_{c_t}}{u_{c_0}} \frac{u_{l_t}}{u_{c_t}} n_t + \sum_{t=0}^{\infty} \beta^t \frac{u_{c_t}}{u_{c_0}} F_{z_t} Z$$
$$+ \{[(1 - \tau_{k_0}) F_{k_0} + 1 - \delta] k_0 + b_0\} = 0$$

- Multiply by $u_c(0)$, define A as before:

$$\sum_{t=0}^{\infty} \beta^t \{u_{ct}c_t - u_{lt}n_t - u_{ct}F_{zt}Z\} - A = 0$$

- Let Φ be Lagrange multiplier for IC and define:

$$V(c_t, n_t, k_t, \Phi) = u(c_t, 1 - n_t) + \Phi(u_{ct}c_t - u_{lt}n_t - u_{ct}F_{zt}Z).$$

- Compare to previous definition of V : Now k_t appears. Why?

- Lagrangian:

$$\mathcal{J} = \sum_{t=0}^{\infty} \beta^t \{ V(c_t, n_t, k_t, \phi) + \theta_t [F(k_t, n_t) - c_t - k_{t+1} + (1 - \delta)k_t - g_t] \} - \phi A$$

- FOC with respect to k_{t+1} :

$$\theta_t = \beta V_{k_{t+1}} + \beta \theta_{t+1} [F_{k_{t+1}} + 1 - \delta]$$

- Compare to FOC in previous model. Now $V_{k_{t+1}}$ appears. Why?

- Suppose there exists a steady state. Therefore:

$$\theta = \beta V_k + \beta \theta [F_k + (1 - \delta)]$$

$$1 = \beta [F_k + (1 - \delta)] + \frac{\beta}{\theta} V_k$$

- Steady state NAC:

$$1 = \beta [(1 - \tau_k) F_k + (1 - \delta)]$$

- Substituting into FOC:

$$\frac{\beta}{\theta} V_k = -\beta \tau_k F_k$$

- Therefore, optimal tax rate is given by:

$$\tau_k = -\frac{V_k}{\theta F_k}$$

- Computing V_k :

$$\frac{\partial V}{\partial k_t} = -\Phi u_{ct} F_{zkt} Z$$

- Hence:

$$\tau_k = \frac{\Phi u_c F_{zk} Z}{\theta F_k}.$$

- Tax rate depends on $Z > 0, u_c > 0, \Phi > 0, \theta > 0, F_k > 0$.
- The sign of τ_k is determined by F_{zk} .
- τ_k is zero only when $F_{zk} = 0$.
- For Cobb-Douglas technology, $F_{zk} > 0 \implies \tau_k > 0$ (**check**).

- We assumed a factor that cannot be taxed.
- In previous model the distortion on the economy was fully transferred to labor, away from capital.
- Here we cannot do this.
- Therefore capital may be taxed.
- How interesting is this example? What is the empirical counterpart of factor Z ?
- In model it is a non-reproducible factor.
- Is it something like land (which is also inelastic)?
- In the real world there are taxes on land ownership.